

problem 3.1: Propositional vs. First-Order Expressiveness

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Part A: Propositional Encoding Cost

In the propositional encoding (as in [x9] and [k7]), adjacency is not encoded as a separate relation—it is embedded in the structure of each biconditional. We count the sentences that explicitly reference which squares neighbor which.

1. **Adjacency:** In propositional logic there is no binary predicate, so adjacency is implicit. If we wanted to *state* it explicitly (e.g., using propositions $A_{(i,j),(i',j')}$), we would need one assertion for every ordered pair of locations: $(n^2)^2 = n^4$ sentences (true for adjacent pairs, false otherwise). For $n = 100$: 10^8 sentences.
2. **Creaking rule:** One biconditional per square, each listing the adjacent D variables in the disjunction. There are n^2 squares, so n^2 sentences. For $n = 100$: 10,000 sentences.
3. **Safety rule:** One biconditional per square: $S_{i,j} \Leftrightarrow \neg D_{i,j} \wedge \neg F_{i,j}$. Again n^2 sentences. For $n = 100$: 10,000 sentences.

Including the rumbling rule (another n^2), the physics encoding requires $3n^2$ biconditionals. For $n = 100$: 30,000 propositional sentences for physics alone.

Part B: FOL Comparison

In first-order logic, the three physics rules each require a single quantified sentence:

1. **Creaking:** $\forall L \text{ Creaking}(L) \Leftrightarrow \exists L' \text{ Adjacent}(L, L') \wedge \text{Damaged}(L')$
2. **Rumbling:** $\forall L \text{ Rumbling}(L) \Leftrightarrow \exists L' \text{ Adjacent}(L, L') \wedge \text{Forklift}(L')$
3. **Safety:** $\forall L \text{ Safe}(L) \Leftrightarrow \neg \text{Damaged}(L) \wedge \neg \text{Forklift}(L)$

3 sentences encode the physics for any grid size. Structural facts still scale: closed-world adjacency requires $O(n^4)$ ground assertions (one per ordered pair of locations, stating adjacent or not), and domain closure adds $O(n^2)$ terms. But the physics rules themselves—the part that grows as $3n^2$ in the propositional encoding—are constant in FOL.

Part C: Translation Challenge

FOL: The statement "there is exactly one damaged floor section" is concise:

$$\exists L \text{ Damaged}(L) \wedge \forall L' (\text{Damaged}(L') \Rightarrow L' = L)$$

Propositional logic for 3×3 : With 9 damage variables $D_{1,1}, D_{2,1}, \dots, D_{3,3}$, we need:

- **At-least-one:** $D_{1,1} \vee D_{2,1} \vee D_{3,1} \vee D_{1,2} \vee D_{2,2} \vee D_{3,2} \vee D_{1,3} \vee D_{2,3} \vee D_{3,3}$ (1 clause)
- **At-most-one:** For every pair of distinct variables, assert that they are not both true: $\neg D_{i,j} \vee \neg D_{i',j'}$. There are $\binom{9}{2} = 36$ such clauses.

Total: $1 + 36 = 37$ clauses for 3×3 .

Scaling: For an $n \times n$ grid with n^2 damage variables, the at-most-one constraint requires $\binom{n^2}{2} = \frac{n^2(n^2-1)}{2}$ pairwise exclusion clauses, which is $O(n^4)$. For $n = 100$: $\binom{10000}{2} = 49,995,000$ clauses. The FOL version remains 1 sentence regardless of n .

So yes, the statement *can* be expressed in propositional logic, but the encoding cost grows as $O(n^4)$ —a dramatic illustration of the expressiveness gap.

Part D: Practical Implications

1. **Propositional at scale:** The basic physics encoding ($3n^2 = 30,000$ biconditionals for $n = 100$) is well within the capacity of modern SAT solvers, which routinely handle millions of clauses. However, richer constraints like "exactly one damaged floor" add $O(n^4) \approx 50$ million clauses, which is feasible but pushing practical limits. Adding multiple such cardinality constraints rapidly becomes unwieldy.
2. **FOL restrictions for tractability:** Unrestricted FOL is semi-decidable (and undecidable for satisfiability in general). Tractable restrictions include: (a) *Horn clauses*, which enable polynomial-time forward and backward chaining; (b) *function-free* FOL over finite domains, which is decidable; (c) *description logics*, which restrict

quantifier patterns to guarantee decidability while retaining useful expressiveness; (d) *bounded-variable fragments*, which limit the number of variables.

3. **Hybrid approach:** Use FOL for the general physics rules (3 quantified sentences, grid-size independent), and *propositionalize* only in the local neighborhood around the agent's current position for fast inference. For example, the agent could maintain a FOL KB for global constraints but ground the creaking/rumbling rules only for the squares within k steps of its current location. This combines FOL's compact representation with propositional logic's efficient (and decidable) inference. Z3 effectively does this internally via MBQI (Model-Based Quantifier Instantiation), grounding quantifiers on demand.

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